Stochastic Contextual Bandits / Part 2

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Linear bandits

(a generalization of finite bandits)

Setting: At each round $t = 1, 2, \ldots,$

0. A decision set $\mathcal{D}_t \subset \mathbb{R}^d$ is revealed
1. Statistician picks $D_t \in \mathcal{D}_t$
2. She gets a reward $Y_t = \langle \theta^\star, D_t \rangle + \eta_t$
   where η_t is an independent noise
3. This reward Y_t is the only feedback she receives

$\longrightarrow$ Exploration–exploitation dilemma
estimate $\theta^\star$ vs. get high rewards Y_t

Goal:
Maximize cumulative rewards $\longleftrightarrow$ Minimize pseudo-regret

$$R_T = \sum_{t=1}^{T} \max_{a \in \mathcal{D}_t} \langle \theta^\star, a \rangle - \sum_{t=1}^{T} \langle \theta^\star, D_t \rangle$$

Pseudo-regret: cf. $\sum_{t=1}^{T} Y_t$ is $\sqrt{T}$ close to $\sum_{t=1}^{T} \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] = \sum_{t=1}^{T} \langle \theta^\star, D_t \rangle$

Setting: pick $D_t \in \mathcal{D}_t \subset \mathbb{R}^d$ and get $Y_t = \underbrace{\langle \theta^\star, D_t \rangle}_{\text{mean effect}} + \underbrace{\eta_t}_{\text{noise}}$

Finite-armed bandits: $d = K$ and $\mathcal{D}_t = \{\mathbf{e}_1, \ldots, \mathbf{e}_K\}$

$(\mathbf{e}_1, \ldots, \mathbf{e}_K)$ is the canonical basis of $\mathbb{R}^K$

The mean rewards equal $\mu_a^\star = \theta_a^\star$

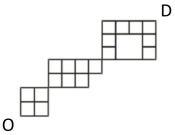
Sleeping finite-armed bandits: $d = K$ and $\mathcal{D}_t \subseteq \{\mathbf{e}_1, \ldots, \mathbf{e}_K\}$

Combinatorial bandits

Selection of m best arms: e.g., d ads, m comparable locations
$\rightarrow \mathcal{D} = \left\{ a \in \{0,1\}^d : \|a\|_1 = m \right\}$

Shortest path: d edges, set of paths from O to D
$\rightarrow \mathcal{D} \subseteq \{0,1\}^d$

Only the total time from O to D is observed



Setting: pick $D_t \in \mathcal{D}_t \subset \mathbb{R}^d$ and get $Y_t = \langle \theta^\star, D_t \rangle + \eta_t$

Linear contextual bandits: At each round $t = 1, 2, \ldots$,

0. A context $\mathbf{x}_t \in \mathbb{R}^m$ is revealed
No assumption on its generation
1. Statistician picks $A_t \in [K]$
2. She gets a reward $Y_t = \big\langle \theta^\star, \, \varphi(\mathbf{x}_t, A_t) \big\rangle + \eta_t$
   where η_t is an independent noise
3. This reward Y_t is the only feedback she receives

Comments

- $\theta^\star \in \mathbb{R}^d$ is unknown
- The transfer function $\varphi : \mathbb{R}^m \times [K] \to \mathbb{R}^d$ is known
- Amounts to picking in $\mathcal{D}_t = \big\{ \varphi(\mathbf{x}_t, a) : a \in [K] \big\}$
- More concrete formulation good for modeling $\to$ stick to it
- Pseudo-regret rewrites as:

$$R_T = \sum_{t \leqslant T} \max_{a \in [K]} \big\langle \theta^\star, \, \varphi(\mathbf{x}_t, a) \big\rangle - \sum_{t \leqslant T} \big\langle \theta^\star, \, \varphi(\mathbf{x}_t, A_t) \big\rangle$$

Setting: context $\mathbf{x}_t$, arm $A_t \in [K]$, reward $Y_t = \langle \theta^\star, \varphi(\mathbf{x}_t, A_t) \rangle + \eta_t$

Pseudo-regret $\displaystyle \sum_{t \leqslant T} \max_{a \in [K]} \langle \theta^\star, \varphi(\mathbf{x}_t, a) \rangle - \sum_{t \leqslant T} \langle \theta^\star, \varphi(\mathbf{x}_t, A_t) \rangle$

Key: learn $\theta^\star$ (= estimate it while playing) — intuition:

$$\sum_{s \leqslant t-1} \varphi(\mathbf{x}_s, A_s) Y_s \approx \sum_{s \leqslant t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^{\mathsf{T}} \theta^\star \quad \text{are } \sqrt{t}\text{–close}$$

Thus $\theta_\star$ close to $\displaystyle \left(\sum_{s \leqslant t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^{\mathsf{T}} \right)^{-1?} \sum_{s \leqslant t-1} \varphi(\mathbf{x}_s, A_s) Y_s$

$\to$ LinUCB with regularization $\lambda > 0$

Abbasi-Yadkori, Pál, Szepesvári [2011]

$$\widehat{\theta}_{t-1} = \left(\lambda \,\mathrm{Id}_d + \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^{\mathsf{T}} \right)^{-1} \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) Y_s$$

Setting: context $\mathbf{x}_t$, arm $A_t \in [K]$, reward $Y_t = \langle \theta^\star, \varphi(\mathbf{x}_t, A_t) \rangle + \eta_t$

Pseudo-regret $\displaystyle \sum_{t \leqslant T} \max_{a \in [K]} \langle \theta^\star, \varphi(\mathbf{x}_t, a) \rangle - \sum_{t \leqslant T} \langle \theta^\star, \varphi(\mathbf{x}_t, A_t) \rangle$

LinUCB with regularization $\lambda > 0$ (Abbasi-Yadkori, Pál, Szepesvári, 2011)

Step 1: estimate

$$\widehat{\theta}_{t-1} = \left(\lambda \, \mathrm{Id}_d + \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^\mathsf{T} \right)^{-1} \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \, Y_s$$

Step 2 [optimistic choice]: pick

$$A_t \in \mathop{\arg\max}_{a \in [K]} \left\{ \varphi(\mathbf{x}_t, a)^\mathsf{T} \, \widehat{\theta}_{t-1} + \varepsilon_{t-1}(\mathbf{x}_t, a) \right\}$$

Confidence bonus $\varepsilon_{t-1}(\mathbf{x}_t, a)$ and regularization $\lambda > 0$ to be determined by the analysis

Reminder: $$\widehat{\theta}_{t-1} = \underbrace{\left(\lambda\,\mathrm{Id}_d + \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s)\varphi(\mathbf{x}_s, A_s)^{\mathsf{T}} \right)^{-1}}_{=G_{t-1}} \underbrace{\sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s)\Bigl(\varphi(\mathbf{x}_s, A_s)^{\mathsf{T}}\,\theta^\star}_{=G_{t-1}-\lambda\,\mathrm{Id}_d} + \eta_s \Bigr)$$

Analysis 1/2: determination of $\varepsilon_{t-1}(\mathbf{x}_t, a)$

Such that $\left| \varphi(\mathbf{x}_t, a)^{\mathsf{T}}\,\widehat{\theta}_{t-1} - \varphi(\mathbf{x}_t, a)^{\mathsf{T}}\theta^\star \right| \leqslant \varepsilon_{t-1}(\mathbf{x}_t, a)$ w.p. $1-\delta$

Proof: $$\widehat{\theta}_{t-1} - \theta^\star = G_{t-1}^{-1}\left(-\lambda\,\mathrm{Id}_d\,\theta^\star + \sum_{s\leqslant t-1} \varphi(\mathbf{x}_s, A_s)\,\eta_s \right)$$ thus

$$\left| \varphi(\mathbf{x}_t, a)^{\mathsf{T}}\,\widehat{\theta}_{t-1} - \varphi(\mathbf{x}_t, a)^{\mathsf{T}}\theta^\star \right| \leqslant \left\| G_{t-1}^{-1/2}\,\varphi(\mathbf{x}_t, a) \right\| \, \left\| G_{t-1}^{1/2}\bigl(\widehat{\theta}_{t-1} - \theta^\star\bigr) \right\|$$

$$\leqslant \underbrace{\left\| G_{t-1}^{-1/2}\,\varphi(\mathbf{x}_t, a) \right\|}_{\text{known}} \left(\sqrt{\lambda}\, \underbrace{\|\theta^\star\|}_{\leqslant C} + \underbrace{\left\| G_{t-1}^{-1/2} \sum_{s\leqslant t-1} \varphi(\mathbf{x}_s, A_s)\,\eta_s \right\|}_{\text{see next slide}} \right)$$

where we used $G_{t-1} \succeq \lambda\,\mathrm{Id}_d$ so that $G_{t-1}^{-1/2}(\lambda \mathrm{Id}_d) \preceq \sqrt{\lambda}\,\mathrm{Id}_d$

Intermezzo: Martingales + Linear algebra

Theorem (Abbasi-Yadkori, Pál, Szepesvári, 2011)

Filtration $(\mathcal{F}_s)_{s \geqslant 0}$

Scalar-valued $(\eta_s)_{s \geqslant 1}$ *adapted & σ^2–sub-Gaussian cond. to* $\mathcal{F}_{s-1}$

d–vector-valued $(X_s)_{s \geqslant 1}$ *predictable &* $\|X_s\| \leqslant 1$

Let $\quad M_t = \sum_{s \leqslant t} \eta_s X_s \quad$ *and* $\quad G_t = \lambda \, \mathrm{Id}_d + \sum_{s \leqslant t} X_s X_s^{\mathsf{T}}$

W.p. $1 - \delta$, *for all* $t \geqslant 1$,

$$\big\| G_t^{-1/2} \, M_t \big\| \leqslant \sigma \sqrt{2 \ln(1/\delta) + d \ln\big(1 + t/(\lambda d)\big)} = \mathcal{O}\big(\sqrt{\ln(t/\lambda)}\big)$$

Application: $X_s = \varphi(\mathbf{x}_s, A_s)$
& cond. to $\mathcal{F}_{s-1}$ $\leftrightarrow$ integr. randomness η_s

Intermezzo: Martingales + Linear algebra

1. Laplace's method for $f : [a, b] \to \mathbb{R}$ with single max. at u_0

$$I_x = \int_a^b \exp\big(x\, f(u)\big)\, \mathrm{d}u = \exp\big(x\, f(u_0)\big) \underbrace{\int_a^b \exp\Big(x\, \big(f(u) - f(u_0)\big)\Big)\, \mathrm{d}u}_{\leqslant \text{ constant}}$$

$$\text{thus} \qquad f(u_0) \approx \frac{\ln I_x}{x} \qquad \text{for large } x > 0$$

2. To be used over the supermartingales (cf. sub-Gaussian noises)

$$E_t(x) = \exp\left(\sum_{s \leqslant t} x^\top X_s\, \eta_s - \frac{\sigma^2}{2}\big(x^\top X_s\big)^2\right) = \exp\Big(x^\top M_t - (\sigma^2/2)\, x^\top G_t x\Big)$$

Intermezzo: Martingales + Linear algebra

Supermartingales $\quad E_t(x) = \exp\Big(x^\top M_t - (\sigma^2/2)\, x^\top G_t x \Big)$

Supermartingale $\quad \overline{E}_t = \displaystyle\int_{\mathbb{R}^d} E_t(x)\,\mathrm{d}x$

$$= [\ldots] = \sqrt{\det\big(2\pi\sigma^{-2}\, G_t^{-1}\big)}\ \exp\left(\frac{1}{2\sigma^2} \big\Vert\, G_t^{-1/2}\, M_t \,\big\Vert^2 \right)$$

Cf. formulas for Gaussian densities

Doob: $\quad \mathbb{E}\big[\,\overline{E}_t\,\big] \leqslant \mathbb{E}\big[\,\overline{E}_0\big] = [\ldots] = \sqrt{\det\big(2\pi\lambda^{-1}\,\mathrm{Id}_d\big)}$

then Markov:

$$\mathbb{P}\Big(\big\Vert\, G_t^{-1/2}\, M_t \,\big\Vert > u \Big) \leqslant \exp\left(-\frac{u^2}{2\sigma^2} \right)\ \mathbb{E}\left[\frac{1}{2\sigma^2} \exp\Big(\big\Vert\, G_t^{-1/2}\, M_t \,\big\Vert^2 \Big) \right]$$

Also bound $\quad \det(G_t) \leqslant (\lambda + t/d)^d$

$$Y_t = \langle \theta^\star,\, \varphi(\mathbf{x}_t, A_t)\rangle + \eta_t \quad \text{and} \quad \widehat{\theta}_{t-1} = \underbrace{\left(\lambda\,\mathrm{Id}_d + \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s)\varphi(\mathbf{x}_s, A_s)^{\mathsf{T}}\right)}_{=G_{t-1}}^{-1} \underbrace{\sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s)\Big(\varphi(\mathbf{x}_s, A_s)^{\mathsf{T}}\,\theta^\star + \eta_s\Big)}_{=G_{t-1} - \lambda\,\mathrm{Id}_d}$$

Analysis 1/2: determination of $\varepsilon_{t-1}(\mathbf{x}_t, a)$

Such that $\left|\varphi(\mathbf{x}_t, a)^{\mathsf{T}}\,\widehat{\theta}_{t-1} - \varphi(\mathbf{x}_t, a)^{\mathsf{T}}\theta^\star\right| \leqslant \varepsilon_{t-1}(\mathbf{x}_t, a)$ w.p. $1-\delta$

Proof: so far, w.p. $1-\delta$,

$$\left|\varphi(\mathbf{x}_t, a)^{\mathsf{T}}\,\widehat{\theta}_{t-1} - \varphi(\mathbf{x}_t, a)^{\mathsf{T}}\theta^\star\right| \leqslant \left\| G_{t-1}^{-1/2}\,\varphi(\mathbf{x}_t, a)\right\| \, \left\| G_{t-1}^{1/2}\big(\widehat{\theta}_{t-1} - \theta^\star\big)\right\|$$

$$\leqslant \underbrace{\left\| G_{t-1}^{-1/2}\,\varphi(\mathbf{x}_t, a)\right\|}_{\text{known}} \left(\sqrt{\lambda}\,\underbrace{\|\theta^\star\|}_{\leqslant C} + \underbrace{\left\| G_{t-1}^{-1/2} \sum_{s \leqslant t-1} \varphi(\mathbf{x}_s, A_s)\,\eta_s \right\|}_{\text{known bound } \mathcal{O}\left(\sqrt{\ln(t/\lambda)}\right)}\right)$$

$\rightarrow$ we read our $\varepsilon_{t-1}(\mathbf{x}_t, a)$

$\widehat{\theta}_{t-1}$ s.t. $\forall a \in [K]$, $\left|\varphi(\mathbf{x}_t, a)^{\mathsf{T}}\,\widehat{\theta}_{t-1} - \varphi(\mathbf{x}_t, a)^{\mathsf{T}}\,\theta^\star\right| \leqslant \varepsilon_{t-1}(\mathbf{x}_t, a) = \left\| G_{t-1}^{-1/2}\,\varphi(\mathbf{x}_t, a)\right\| \left(C\sqrt{\lambda} + \mathcal{O}\left(\sqrt{\ln(t/\lambda)}\right)\right)$

$\longrightarrow$ Pick $A_t \in \operatorname*{arg\,max}_{a\in[K]} \left\{\varphi(\mathbf{x}_t, a)^{\mathsf{T}}\,\widehat{\theta}_{t-1} + \varepsilon_{t-1}(\mathbf{x}_t, a)\right\}$ and get $Y_t = \langle \theta^\star, \varphi(\mathbf{x}_t, A_t)\rangle + \eta_t$

Aim: control pseudo-regret $\displaystyle\sum_{t\leqslant T} \max_{a\in[K]} \varphi(\mathbf{x}_t, a)^\top \theta^\star - \sum_{t\leqslant T} \varphi(\mathbf{x}_t, A_t)^\top \theta^\star$

Analysis 2/2: $\sqrt{T}$–gap-free regret bound

Proof: $-\varphi(\mathbf{x}_t, A_t)^{\mathsf{T}}\theta^\star \leqslant \varepsilon_{t-1}(\mathbf{x}_t, A_t) - \varphi(\mathbf{x}_t, A_t)^{\mathsf{T}}\,\widehat{\theta}_{t-1}$ and

$$\max_{a\in[K]} \varphi(\mathbf{x}_t, a)^{\mathsf{T}}\theta^\star \leqslant \max_{a\in[K]} \left\{\varphi(\mathbf{x}_t, a)^{\mathsf{T}}\widehat{\theta}_{t-1} + \varepsilon_{t-1}(\mathbf{x}_t, a)\right\}$$

$$= \varphi(\mathbf{x}_t, A_t)^{\mathsf{T}}\widehat{\theta}_{t-1} + \varepsilon_{t-1}(\mathbf{x}_t, A_t) \qquad \text{cf. def. } A_t$$

Thus,

$$\sum_{t\leqslant T} \max_{a\in[K]} \varphi(\mathbf{x}_t, a)^\top \theta^\star - \sum_{t\leqslant T} \varphi(\mathbf{x}_t, A_t)^\top \theta^\star \leqslant \sum_{t\leqslant T} \varepsilon_{t-1}(\mathbf{x}_t, A_t)$$

$$\leqslant \underbrace{\left(C\sqrt{\lambda} + \mathcal{O}\left(\sqrt{\ln(T/\lambda)}\right)\right)}_{\leqslant \widetilde{\mathcal{O}}(1)} \underbrace{\sum_{t\leqslant T} \left\| G_{t-1}^{-1/2}\,\varphi(\mathbf{x}_t, A_t)\right\|}_{\leqslant \widetilde{\mathcal{O}}\left(\sqrt{T}\right) \quad \text{see next slide}}$$

Remains to show $$\sum_{t \leqslant T} \left\| G_{t-1}^{-1/2}\, \varphi(\mathbf{x}_t, A_t) \right\| \leqslant \widetilde{\mathcal{O}}\big(\sqrt{T}\big)$$

Dealing with $\varphi(\mathbf{x}_t, A_t)$ is vital!

Elliptic potential lemma (Abbasi-Yadkori, Pál, Szepesvári, 2011)

$$G_{t-1} = \overbrace{\lambda\, \mathrm{Id}_d + \sum_{s \leqslant t-1} X_s X_s^{\mathsf{T}}}^{\succeq \lambda\, \mathrm{Id}_d \succeq \mathrm{Id}_d} \qquad \text{where} \qquad X_s = \varphi(\mathbf{x}_s, A_s)$$

$$\sum_{t \leqslant T} \left\| G_{t-1}^{-1/2}\, X_t \right\| \leqslant \sqrt{T \sum_{t \leqslant T} X_t^{\top} G_{t-1}\, X_t} \qquad \text{C-Schw. } \& \ X_t^{\top} G_{t-1}\, X_t \leqslant \|X_t\|^2 \leqslant 1$$

$$\leqslant \sqrt{2T \sum_{t \leqslant T} \ln\Big(1 + X_t^{\top} G_{t-1}\, X_t\Big)} \qquad u \leqslant \ln(1+u) \text{ for } u \leqslant 1$$

$$= \sqrt{2T \frac{\ln\big(\det(G_T)\big)}{\ln\big(\det(\lambda\, \mathrm{Id}_d)\big)}} \qquad \text{equality: "matrix-determinant lemma"}$$

$$\leqslant \sqrt{2T\ \widetilde{\mathcal{O}}(1)}$$

Linear contextual bandits: optimality

See Lattimore and Szepesvári [2020, Chapter 25]

The $\widetilde{\mathcal{O}}(\sqrt{T})$ gap-free bound exhibited for LinUCB is fine

In T gap-dependent bounds may be achieved with a different class of algorithms
- not based on optimism
- performing tracking à la Graves and Lai [1997]

Logistic bandits

Extended from Faury, Abeille, Calauzènes, Fercoq [2020] See GLMs in Filippi, Cappé,

Garivier, Szepesvári [2010]

At each round $t = 1, 2, \ldots,$

0. A context $\mathbf{x}_t \in \mathbb{R}^d$ is determined by the environment
1. Statistician picks arm $A_t \in [K]$
2. The outcome $Y_t \in \{0, 1\}$ is drawn with probability $P(\mathbf{x}_t, A_t)$
   This is the only feedback Statistician receives
3. Statistician gets the reward $r(\mathbf{x}_t, A_t)\, Y_t$

Conversion rate P unknown but reward function r known

Structural assumption:

$$P(\mathbf{x}, a) = \eta\big(\varphi(\mathbf{x}, a)^{\mathsf{T}} \theta_\star\big) \qquad \text{where} \qquad \eta(x) = \frac{1}{1 + \mathrm{e}^{-x}}$$

Similar results may be achieved as for linear bandits

Estimation based on maximum likelihood

Linear contextual bandits with latent spates

See Nelson et al. [2022], Li and Stoltz [2026]

Latent states $\qquad h_t \in [H] \qquad$ e.g., economic state

Rewards $\qquad Y_t = \varphi(\mathbf{x}_t, A_t)^\top \theta^\star_{h_t} + \eta_t \qquad \rightarrow \qquad (\theta^\star_h)_{h\in[H]} \in \mathbb{R}^{dH}$

Hidden Markov model: $\widehat{\mathbf{b}}_t(h)$ estimates $\mathbf{b}_t(h) = \mathbb{P}\big(h_t = h \mid \mathbf{x}_{1:t}\big)$

LinUCB strategy on estimated beliefs:

- Estimate $\qquad \big(\widehat{\theta}_{t-1,h}\big)_{h\in[H]} = G_{t-1}^{-1} \displaystyle\sum_{s=1}^{t-1} \big(\widehat{\mathbf{b}}_s(h)\varphi(\mathbf{x}_s, A_s)\big)_{h\in[H]} Y_s$
- Pick
  $$A_t \in \operatorname*{arg\,max}_{a\in[K]} \left\{ \sum_{h\in[H]} \widehat{\mathbf{b}}_t(h)\, \varphi(\mathbf{x}_t, a)^\top\, \widehat{\theta}_{\ldots,h} + \varepsilon_{t-1}(\mathbf{x}_t, a) \right\}$$

Complex statistical dependencies: $\quad A_t$ and $(h_s)_{s\leqslant t}$ are linked through the $(Y_s)_{s\leqslant t}$

$\rightarrow$ HMM forgetting conditions $\quad + \quad$ Proceed in stages: $\widehat{\theta}_{s_t-1,h}$